

Lesson Plan — Andres Cabrera

Date: TBD (first session) **Session length:** 60 minutes (*assumed — not yet confirmed*) **Skill(s):** Desmos orientation (no content skill yet — first exposure, and no diagnostic data exists yet either)

Objectives

- Get comfortable with the SAT's built-in Desmos panel: typing expressions, adding/deleting lines, reading points.
- Learn the universal moves: solve-by-graphing-and-clicking-the-intersection, and sliders for unknown constants.
- Leave the session knowing this is a 25-day sprint to 2026-10-03 — pace and focus matter more than usual.

Warm-up (5 min)

No previous skill exists yet, and there's no diagnostic data (no practice test taken). Skip retrieval warm-up; use this time instead to confirm logistics with Andres (what he's studied, comfort level with algebra generally) since his profile is still mostly blank.

Direct instruction (15 min)

Link: <reference/desmos-playbook.md>

- Show the split-screen layout: add a line, delete a line, zoom/pan.
- Universal move: solve any equation by graphing both sides ($y_1=$, $y_2=$) and clicking the intersection.
- Universal move: sliders for unknown constants.
- Say explicitly: no saving between questions, no image import — don't build habits that assume otherwise.

Desmos move demo

Type $y_1=3x-4$ and $y_2=11$ on two lines, click the intersection, read $x=5$. Then add a slider: type $y_1=a*x-4$, click "add slider" on a , drag it and watch the line rotate.

Guided practice (25 min)

Content-free Desmos fluency drill (no bank dependency — appropriate since there's no skill assignment yet):

1. Graph $y=2x+3$ and click where it crosses the y-axis.
2. Graph $y=x^2-9$ and click both x-intercepts.
3. Solve $4x-7=2x+5$ by graphing both sides and clicking the intersection.
4. Build a table (x1: 1,2,3,4; y1: 5,8,11,14) and type $y1 \sim mx1+b$ to get slope and intercept from regression.
5. Type $(x-1)^2+(y+2)^2=16$ and identify the center and radius by eye.

Exit ticket (10 min)

1. "Solve $5x - 3 = 2x + 9$ using Desmos. What did you type, and what's x?"
2. "Name one thing Desmos on the SAT can't do that surprised you."

Homework assigned

None this session. First content homework goes out once a first skill is picked — either from real practice-test data (preferred, once uploaded) or, if we need to start before that lands, from Algebra fundamentals as a safe default given it's the highest-weight domain.

Notes for me after the session

- Desmos comfort level after this session:
 - General algebra comfort observed (since nothing is documented yet):
 - Still need: grade, courses completed, target score, class days/times/length, homework hours/week.
 - Push for a practice test ASAP — 25 days is tight and we're currently flying blind on skill priority.
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Timing check: $5 + 15 + 25 + 10 = 55$ minutes (assumes ~5 min buffer within a 60-minute session; adjust once real session length is confirmed).